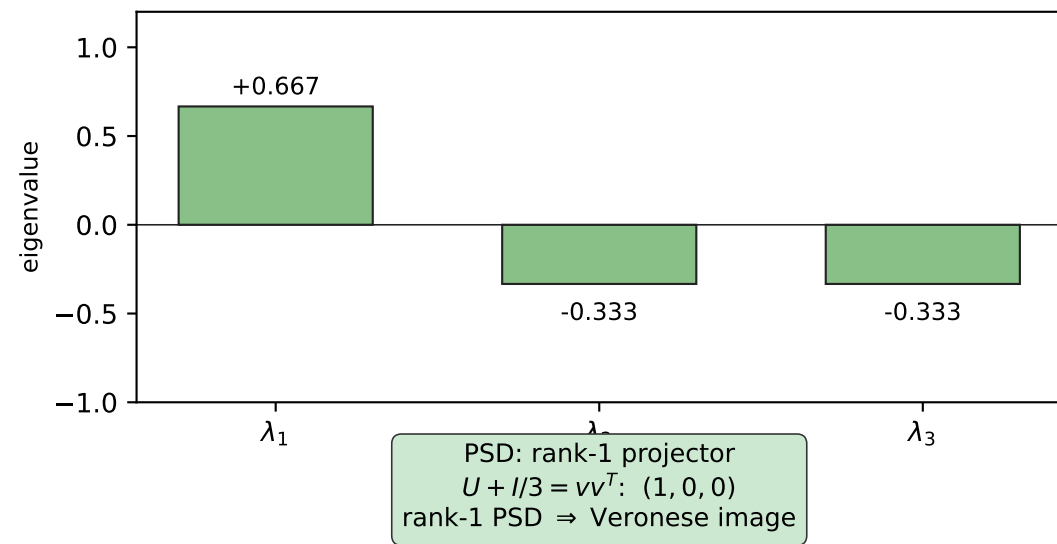
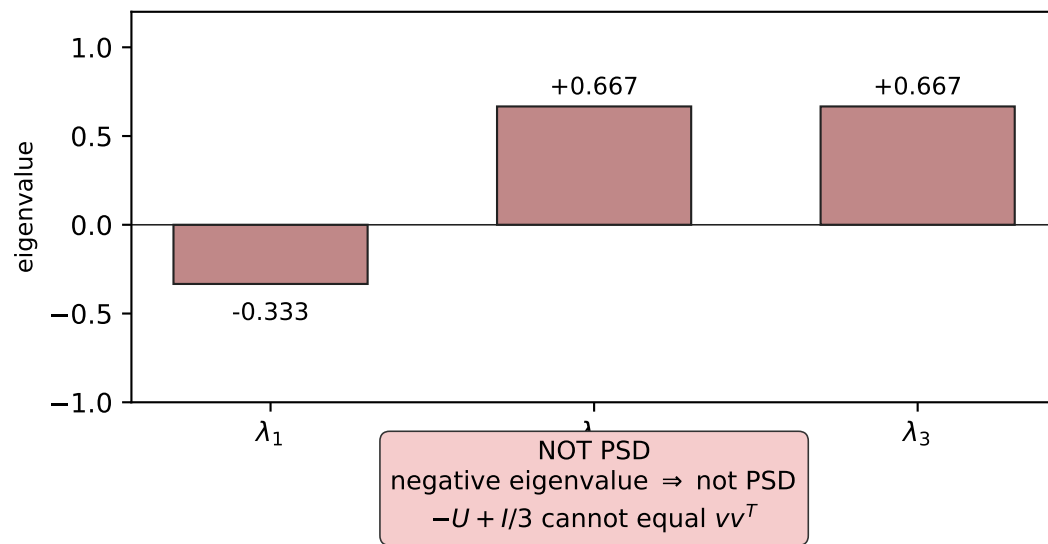


Veronese descent / Schur obstruction (Lemma 3.1, Lemma 4.2, Theorem 4.3)

(A) $U = vv^T - I/3 \in v_2(\mathbb{RP}^2)$
 $\text{spec}(U) = (2/3, -1/3, -1/3)$



(B) $-U \notin v_2(\mathbb{RP}^2)$
 $\text{spec}(-U + I/3) = (-1/3, 2/3, 2/3)$



Schur (Lemma 4.1): an A_5 -equivariant isometry $T: \mathbb{R}^5 \rightarrow \mathbb{R}^5$ is $\pm I$. The case $T = -I$ violates (B): only $T = +I$ survives, giving $O(3)$ -rigidity (Theorem 4.3).